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Cerebro

Your Own Personal Private AI Assistant

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Wearable Intelligence

The 24/7 Context-aware AI Assistant



Cerebro

PRESENTED BY
The Visionaries

Problem Statement

phones have become
a tool that we
use almost from birth.
And we use it a lot for everything.

- These interruptions disrupt cognitive flow and slow task completion in real-time situations.
- Accessing information requires multiple steps (handling the phone, unlocking, app navigation), creating information friction.
- Frequent attention shifts reduce situational awareness, increasing risk during navigation, driving, or social interaction.
- Simple real-time tasks such as navigation, translation, or communication demand multiple attention-shifting steps
- it can sometimes be annoying and even dangerous.
- The screen-centric model is inefficient for time-sensitive tasks and lacks contextual awareness.



UNDERSTANDING THEIR PURPOSE AND FUNCTIONALITY

Smart glasses offer hands-free assistance and privacy.

Our Solution

A hands-free, context-aware alternative is needed to minimize mental effort, enable seamless information access and near instant tool use.

We are not replacing the phone; we are unloading the bloated functionality

- Unlike standard AR glasses that react only to explicit user commands, our system is agentic.
- The glasses continuously understand context, intent, and environment before the user asks.
- They proactively deliver relevant information, guidance, or actions at the right moment.
- This reduces cognitive load, eliminates information friction, and enables natural, hands-free interaction.

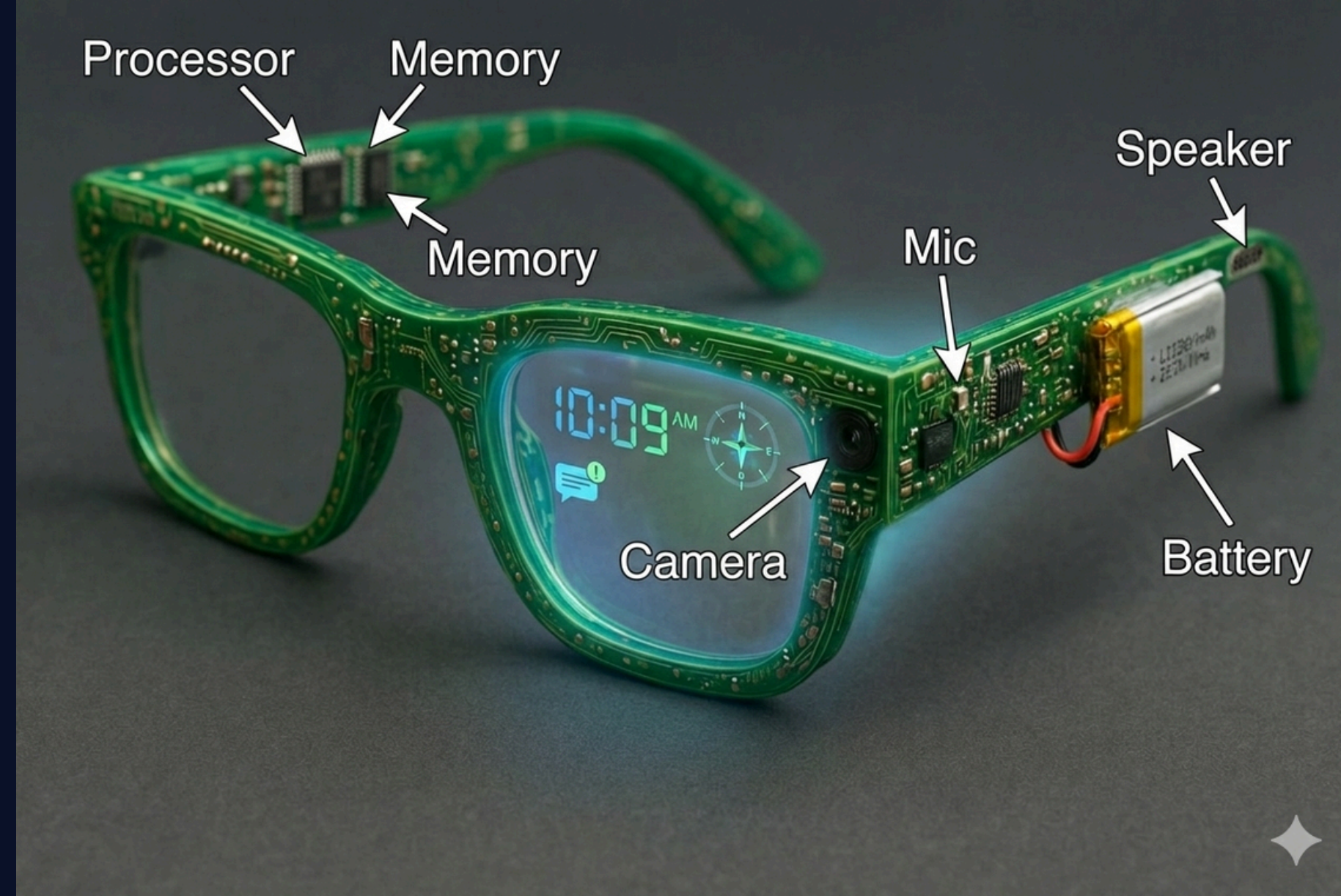


What we need

Are parts available in every phone like:

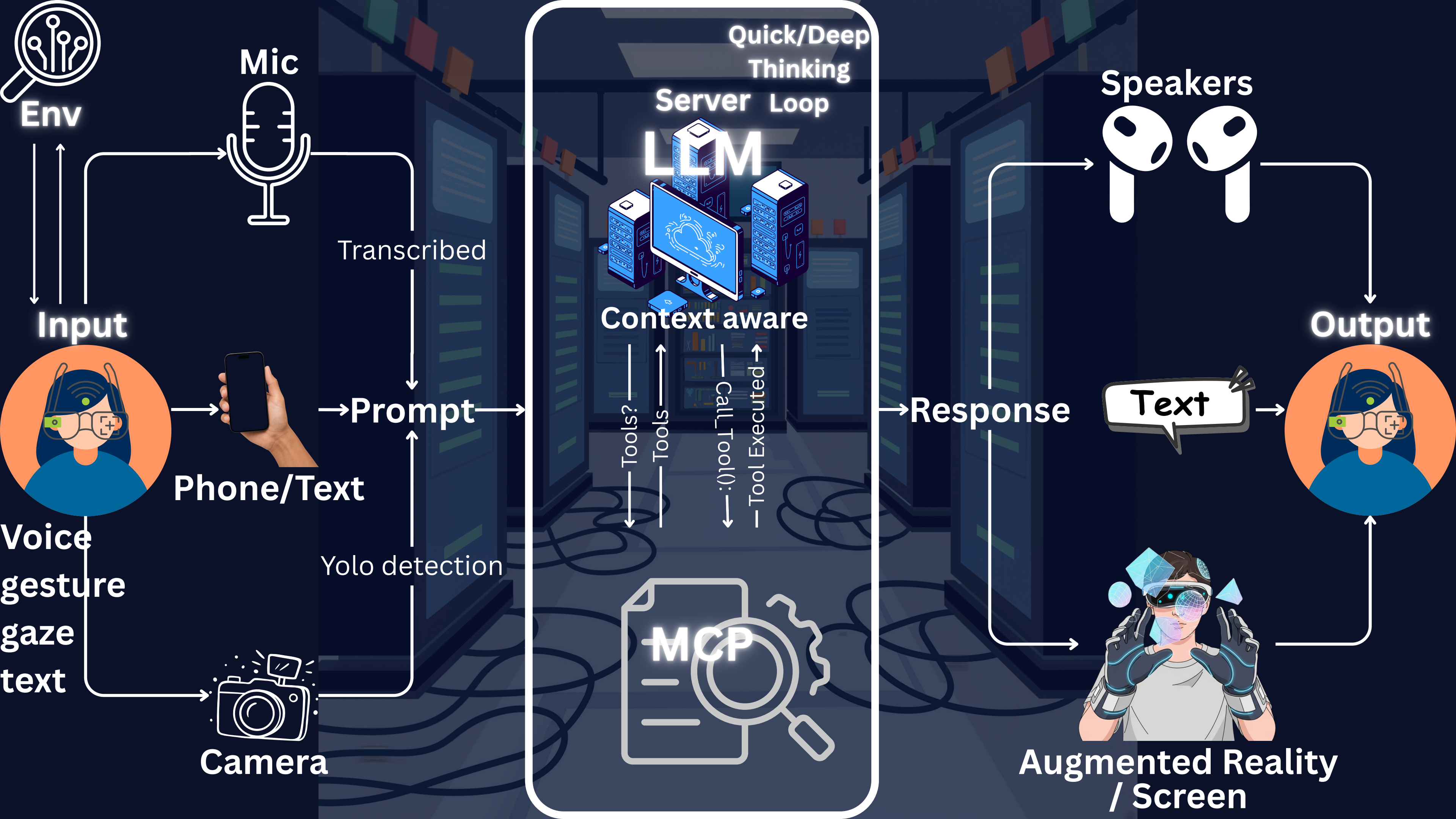
- camera
- mic
- speakers
- gyroscope + imu
- etc...

and connect them all together
then take the data and (for now)
outsource it to our servers



PRIVACY ASSURANCE

Offline intelligence guarantees user **privacy by processing data locally**, minimizing the risk of unauthorized access while ensuring secure interactions without reliance on external cloud services.



Market Analysis & CompetitorsA comparison table

Feature	Our Smart Glasses	Hololens	Meta Ray-Ban	Amazone Amelia
Daily Comfort	High	Low	High	High
Display	HUD+AR	Full MR	None	None
Object Detection	Yes(built-in)	With Customization	Baisc	No
Indoor Navigation	yes	No	No	No
Translation	Real-time	No	No	No
Social AI	Yes	Very limited	No	No
On-Device Processing	Yes	Yes	Limited	Limited
Target Use	Evertday AI assistant +AR Navigation	Industrial MR	Lifestyle	Voice Assistant

User Flow – How the User Interacts with the System

- The user wears the smart glasses during daily activities such as navigation or work.
- The user interacts naturally using voice, gestures, gaze, or head movement without using a phone.
- Built-in sensors (camera, microphone, and IMU) continuously capture environmental and user context in real time.
- The AI agent analyzes the captured data to understand the user's intent and surroundings.
- The decision engine selects the most relevant assistance based on the current context.
- The system delivers hands-free feedback through audio, visual overlays, or notifications in real time.



User Flow Diagram



Business Model Canvas (BMC)

- Key Partners
- Key Activities
- Key Resources
- Value Proposition
- Customer Relationships
- Channels
- Customer Segments
- Cost Structure
- Revenue Streams



Business Model – Value & Customers

Value Proposition

- Hands-free, context-aware AI assistance in real time
- Reduces cognitive load and improves safety and productivity

Customer Segments

- Industrial workers and field professionals
- Visually impaired users and accessibility-focused groups
- Tech-savvy users in healthcare, education, and mobility

Business Model – Market Access & Relationships

Channels

- Direct online sales and enterprise partnerships
- Mobile companion app and authorized distributors

Customer Relationships

- Personalized AI-driven experiences
- Subscriptions, continuous updates, and enterprise support

Business Model – Operations & Revenue

Key Partners

- Hardware manufacturers and AI infrastructure providers
- Industrial, accessibility, and IoT partners

Key Activities

- AI model development and system integration
- Continuous optimization and learning

Key Resources

- Agentic AI system and smart glasses platform
- Context-aware models and datasets

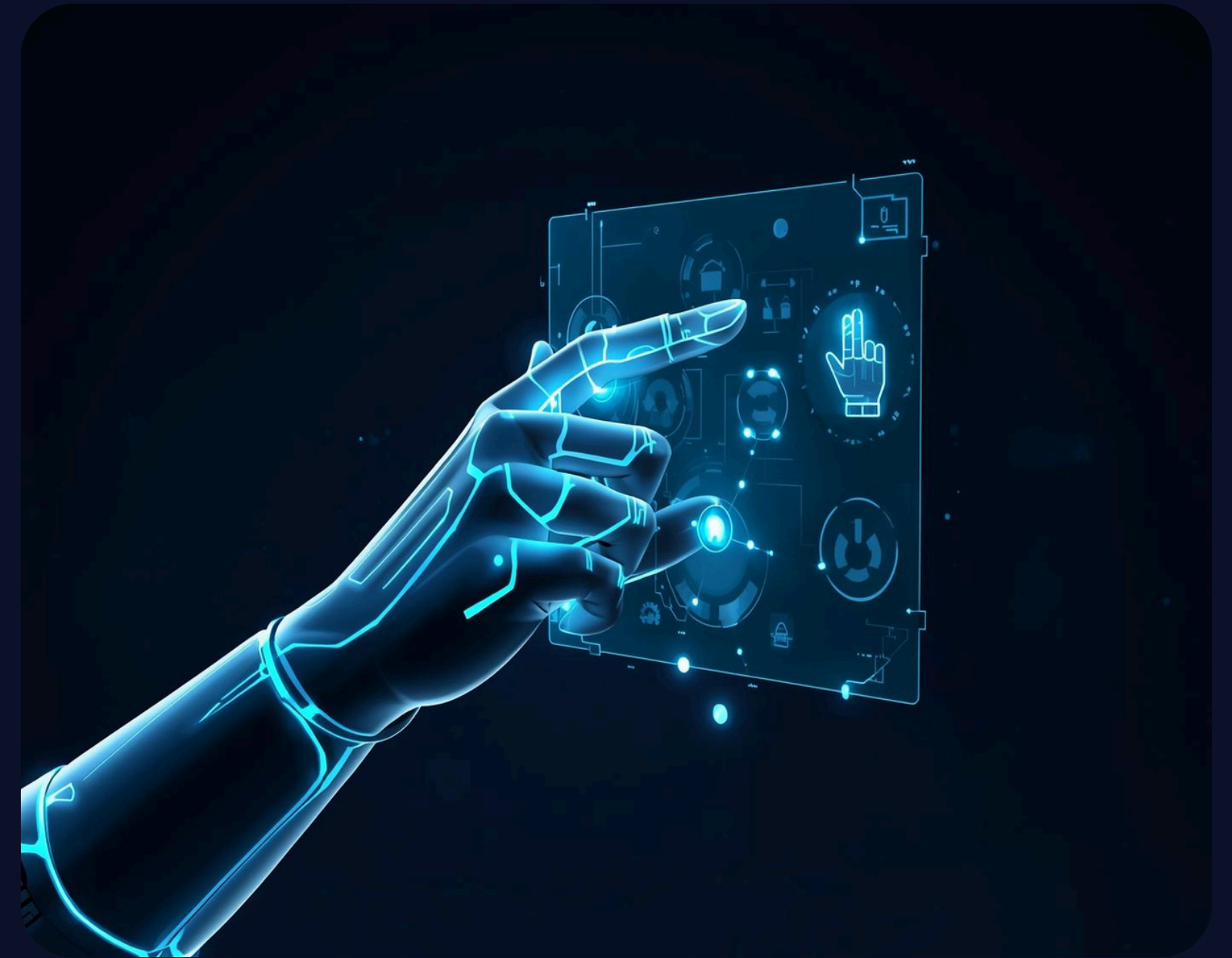
Cost Structure

- Hardware production and AI/cloud costs
- Development, maintenance, and support

Discussion & Key Insights

Strengths

- Hands-free and context-aware interaction improves user focus and safety.
- Agentic AI enables proactive assistance rather than command-based interaction.
- Privacy-first and partial offline processing reduce data dependency and latency.



Limitations

- Hardware constraints such as battery life and processing power.
- Model performance may vary across complex real-world environments.
- Large-scale user testing was limited due to time and resource constraints.

Conclusion

This project presented a smart glasses system powered by Agentic AI to provide hands-free, context-aware assistance in real time.

The system integrates computer vision, speech processing, and contextual sensing to proactively support users without manual interaction.

The proposed solution enhances safety, productivity, and accessibility, especially in industrial and navigation-intensive environments.

By prioritizing privacy and offline intelligence, the system addresses key limitations of existing smart wearable solutions.



Thank you!